Project: Worked examples
Title: Thermal expansion of a restrained bar
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

A bar heated or cooled between two restraints: its free movement, the part closed by a gap before the restraint engages, the restraint force for a rigid or an elastic restraint, the axial stress that results and the movement that actually occurs, with the stress against an allowable.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Free expansion of the bar | $\delta_{free} = \alpha \cdot L \cdot \Delta T$ |
| Expansion actually restrained, once any gap g has closed | $\delta_e = \text{sign}(\delta_{free}) \cdot \max(\lvert\delta_{free}\rvert - g, 0)$ |
| Axial stiffness of the bar | $k_{bar} = E \cdot A / L$ |
| Restraint force, from rigid supports or an elastic restraint $k_s$ | $F = \delta_e \cdot k_{bar}$ (rigid); $F = \delta_e / (1/k_{bar} + 1/k_s)$ (elastic) |
| Thermal stress in the bar, compressive on heating | $\sigma = -F/A$ |
| Net movement of the bar end | $\delta = \delta_{free} - F/k_{bar}$ |

Basis: Thermal stresses[1](Ch. 15); Thermal stresses and strains[2](§3-17).

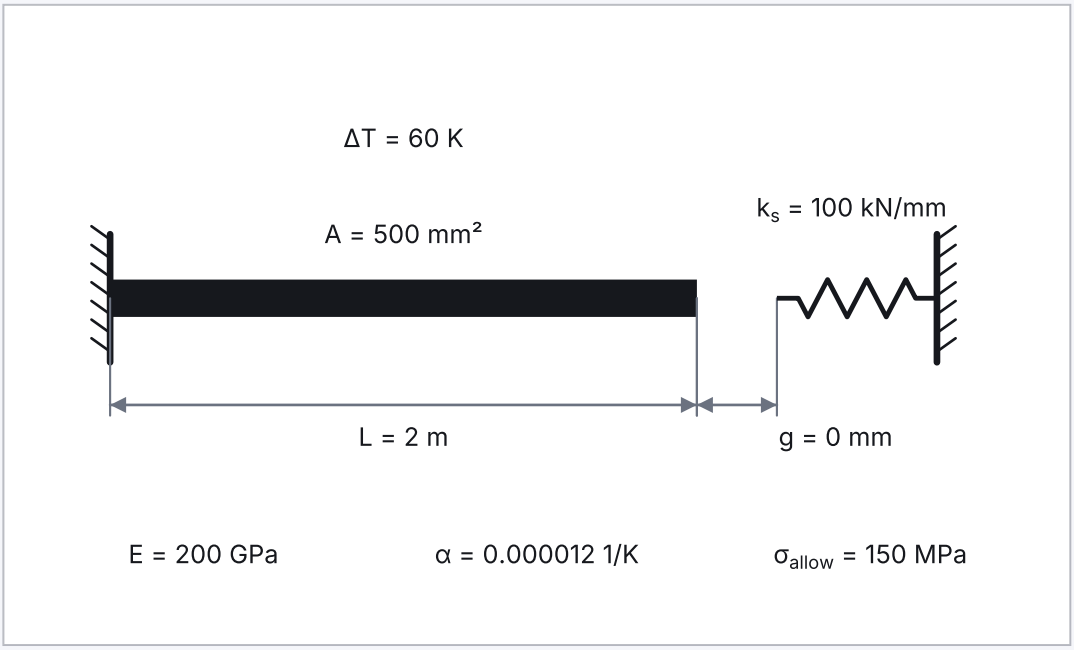



Figure 1 — A bar fixed at one end, a gap, then an elastic restraint of stiffness $k_s$. Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| L | Bar length | 2 | m | drawing: DRG-TX-502 rev A | 0.01 … 1000 m |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| A | Cross-section area | 500 | mm² | drawing: DRG-TX-502 rev A | 1 … 1000000 mm² |
| E | Modulus of elasticity | 200 | GPa | drawing: DRG-TX-502 rev A | 1 … 500 GPa |
| α | Coefficient of thermal expansion | 0.000012 | 1/K | drawing: DRG-TX-502 rev A | 1e-7 … 0.0001 1/K |
| dT | Temperature change (positive heating) | 60 | K | spec: LOAD-SCHEDULE-01 | -300 … 1000 K |
| g | Gap closed before the restraint engages | 0 | mm | drawing: DRG-TX-502 rev A | 0 … 1000 mm |
| $k_s$ | Restraint stiffness (leave empty for a rigid restraint) | 100 | kN/mm | spec: SPEC-MECH-03 design criteria | 0.001 … 10000000 kN/mm |
| $\sigma_{allow}$ | Allowable stress | 150 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |

## 3. Assumptions

A1. The expansion coefficient is constant over the temperature change and the bar stays elastic. (Linear thermoelasticity.)
A2. The bar is uniformly heated and does not buckle under the compressive force. (Check buckling separately (struct.column-buckling).)

## 4. Working

[9] $\delta_{free} = \alpha * L * dT$ [2](§3-17)
= 0.000012 1/K × 2 m × 60 K
= 1.44 mm
Free thermal movement (positive on heating)

[10] $\delta_e = \delta_{free} - g$
= 1.44 mm − 0 mm
= 1.44 mm
Movement after the gap closes (engaged movement)

[11] $k_{bar} = E * A / L$
= 200 GPa × 500 mm² / 2 m
= 50 kN/mm
Axial stiffness of the bar

[12] $F = \delta_e / (1 / k_{bar} + 1 / k_s)$ [1](Ch. 15)
= 1.44 mm / (1 / 50 kN/mm + 1 / 100 kN/mm)
= 48000 N
Restraint force (elastic restraint in series with the bar)

[13] $\sigma = -F / A$
= −48000 N / 500 mm²
= -96 MPa
Axial stress (negative compressive)

[14] $\delta = \delta_{free} - F / k_{bar}$
= 1.44 mm − 48000 N / 50 kN/mm
= 0.48 mm
Actual movement

[15] $u = \mathrm{abs}(\sigma) / \sigma_{allow}$
$= \mathrm{abs}(-96\ \mathrm{MPa}) / 150\ \mathrm{MPa}$
$= 0.64$
Stress utilisation

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| $\delta_{free}$ | 1.44 mm | Free movement |
| $\delta_e$ | 1.44 mm | Engaged movement |
| $k_{bar}$ | 50 kN/mm | Bar stiffness |
| F | 48000 N | Restraint force |
| $\sigma$ | -96 MPa | Axial stress |
| $\delta$ | 0.48 mm | Actual movement |
| u | 0.64 | Stress utilisation |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Thermal stress against allowable | u = 0.64 | u ≤ 1 | 0.64 | |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.thermal-expansion@1.0.1 Content hash 07d40fe119d715ba Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.